

Marketing + Finance Analytics

Executive Report: Channel Profitability, Customer Value, and a Quantified Growth Plan

Prepared for: Executive Leadership Team

Prepared by: Business Analytics Team

Reporting Period: January 2024 - December 2025

Executive Summary

Across the two-year reporting window, the marketing program generated \$1,291,170 in attributed revenue against \$201,634 in spend, converting to an estimated \$1,134,157 in net profit after marketing cost. Blended portfolio efficiency stands at a 6.66:1 LTV:CAC ratio and a 6.40x ROAS.

Three findings drive this report's recommendations: (1) the channel that ranks highest on revenue is not the channel that ranks highest on true net profit once marketing cost and category margins are applied; (2) customer retention in the first 30 days after acquisition is the single largest controllable lever on lifetime value; and (3) a statistically validated creative improvement and a quantified budget reallocation are both ready for immediate execution.

A 15%-of-spend reallocation from the weakest to the strongest elastic paid channels is modeled to shift \$9,185 in spend and generate an estimated \$42,229 in incremental net profit (Section: Strategic Recommendations).

Business Problem

The company markets across seven channels (Google Ads, Facebook, Instagram, Email, Affiliate, TikTok, and Organic) in five countries, but lacked a unified, profit-based view of channel performance. Prior to this engagement, budget decisions were made primarily against top-line revenue and campaign-reported ROAS -- metrics that do not account for true customer lifetime value, retention, or category-level profit margins.

Leadership asked for a data-driven diagnostic answering four questions:

- Which channels and campaigns are actually profitable, not just high-revenue?
- Which customers and acquisition channels produce the most loyal, highest-value customers?
- Is the current marketing budget allocated efficiently?
- What should we expect revenue to do next quarter, and how confident can we be in that number?

Methodology

This analysis follows a six-stage process, each stage documented in a companion Jupyter notebook and a parallel SQL analysis file:

Data Overview & Quality Audit. Ten targeted data-quality checks (missing values, duplicates, invalid dates, negative spend/revenue, orphan records, impossible rates) before any KPI is trusted.

Marketing Performance. Funnel, campaign, channel, device, and country efficiency analysis (CTR, CVR, CPC, CPA, ROAS).

Customer Analytics. RFM segmentation, monthly cohort retention analysis, and customer lifetime value profiling.

Financial Analysis. CAC, LTV, ROI, contribution margin, true channel profitability, and a quantified budget-reallocation model.

Advanced Analytics. A/B test statistical significance testing and revenue forecasting (moving average, linear trend, ARIMA), backtested for accuracy.

Executive Synthesis. KPI dashboard and prioritized, quantified strategic recommendations.

Tooling: Python (pandas, NumPy, Matplotlib, Seaborn, Plotly, statsmodels, SciPy) for analysis and visualization; SQLite + SQL for the relational query analysis (sql/business_queries.sql); all data is synthetic, seeded, and fully reproducible (src/generate_data.py).

Key Performance Indicators

Metric	Value
Total Revenue	\$1,291,170
Marketing Spend	\$201,634
Net Profit	\$1,134,157
Blended CAC	\$33.61
Blended LTV	\$223.71
LTV : CAC Ratio	6.66 : 1
ROAS	6.40x
ROI	562.5%
Conversion Rate	4.17%
Repeat Purchase Rate	83.7%
Next-Quarter Revenue Forecast	\$64,346

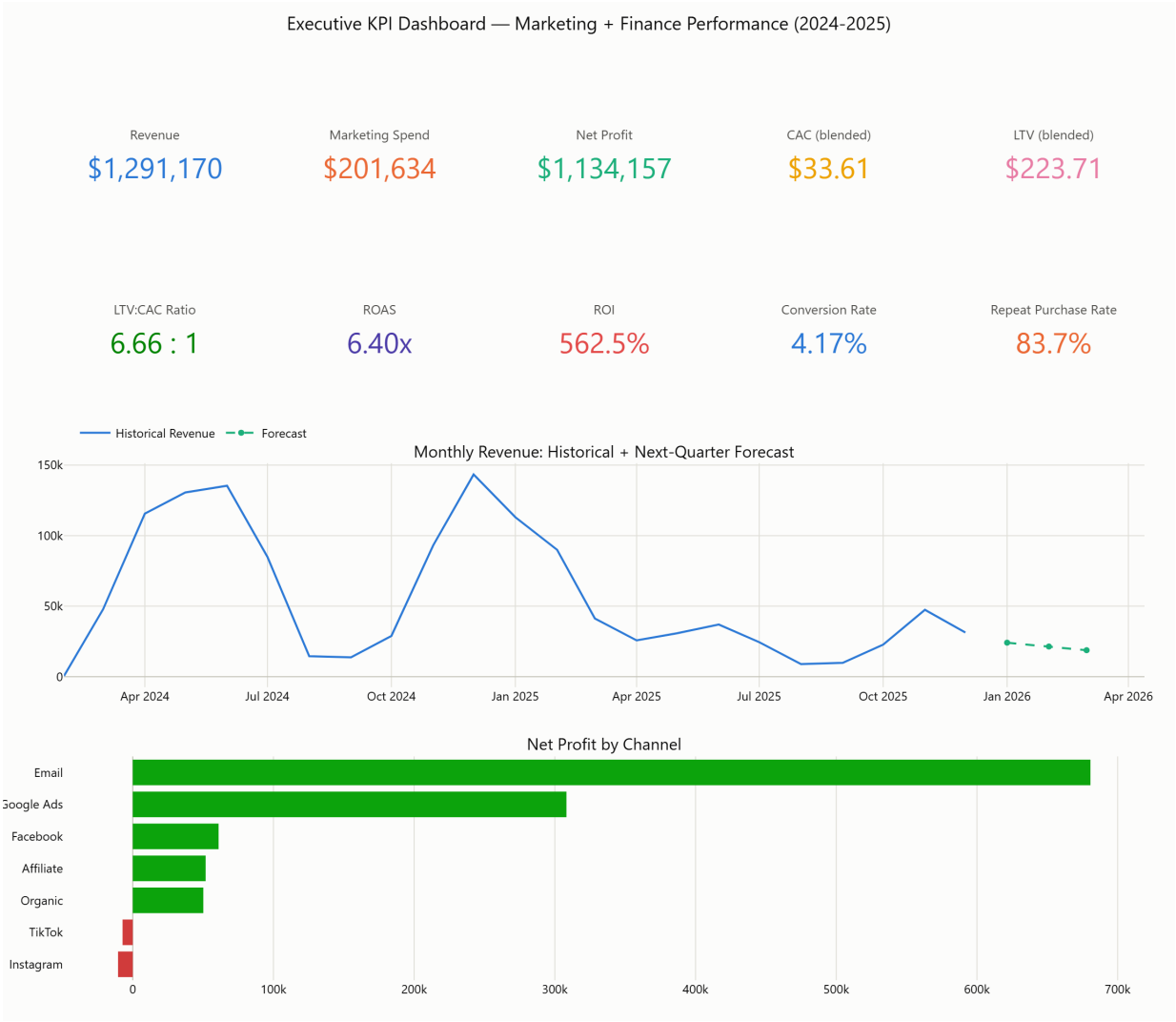


Figure: Executive KPI dashboard (interactive version in notebooks/06_Executive_Report.ipynb).

Major Insights — Marketing Performance

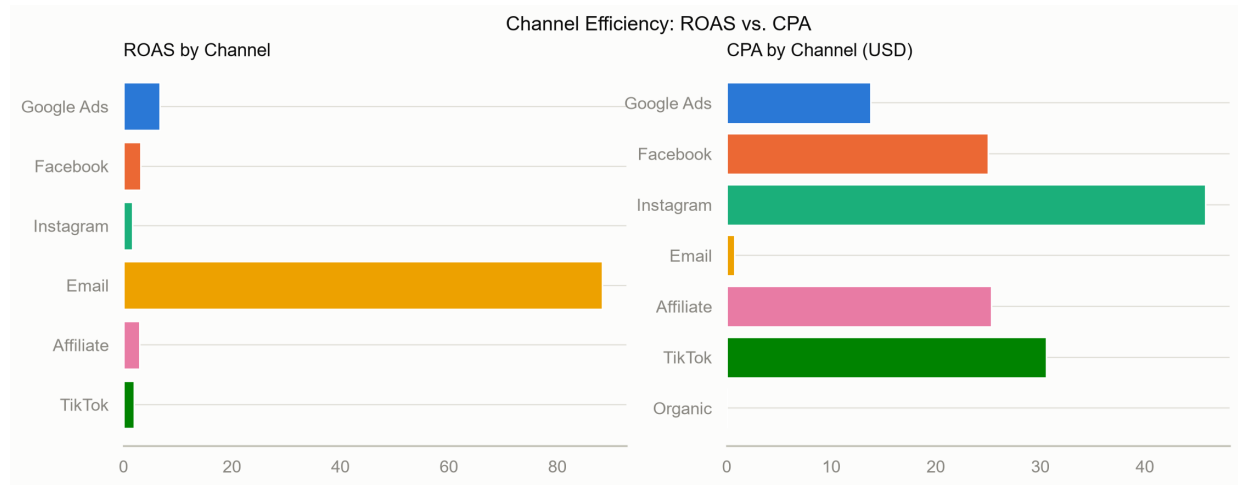


Figure: ROAS and CPA by channel.

Email delivers the strongest ROAS and lowest CPA in the portfolio, reflecting its near-zero media cost against a highly engaged owned audience. Among paid-social channels, TikTok -- the newest, lowest-intent channel -- shows the weakest ROAS. Desktop converts meaningfully better than Mobile, and India combines the lowest CPC with a competitive ROAS, making it the most cost-efficient market to scale.

Major Insights — Customer Analytics



Figure: Monthly cohort retention heatmap.

RFM segmentation identifies a small but disproportionately valuable Champions segment and an At-Risk segment with proven high-frequency purchase history -- the highest-ROI re-engagement target since reactivating a proven buyer is cheaper than acquiring a new one. Cohort analysis shows retention decays most sharply in the first month post-acquisition, and Email/Organic-acquired customers retain meaningfully longer than TikTok/Instagram-acquired customers.

Major Insights — Financial Analysis

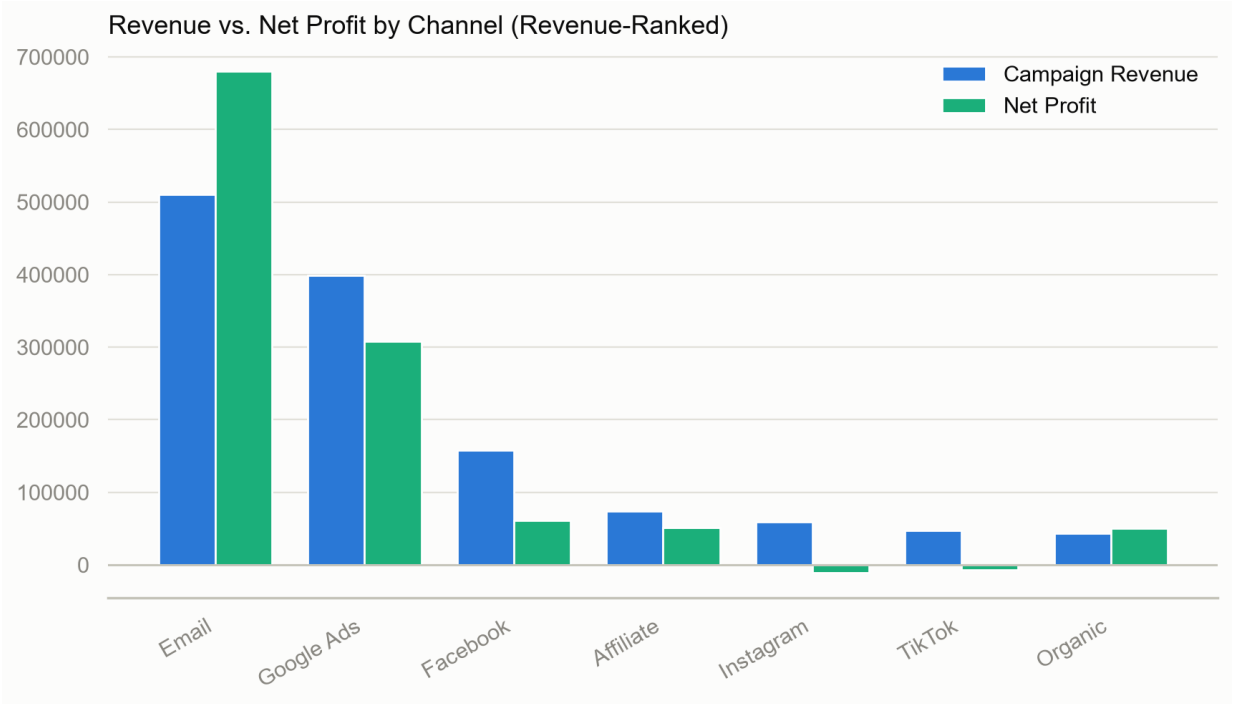


Figure: Campaign revenue vs. net profit by channel.

Email ranks highest on campaign-reported revenue, but **Email** ranks highest on true net profit once marketing spend and category margins are applied. This is the central financial finding of the engagement: a revenue-only leaderboard would misdirect budget.

Channel	Revenue	Spend	Net Profit	Rev. Rank	Profit Rank
Email	\$510,351	\$5,781	\$680,672	1	1
Google Ads	\$398,678	\$59,290	\$308,246	2	2
Facebook	\$157,832	\$49,984	\$61,001	3	3
Affiliate	\$74,484	\$25,348	\$51,795	4	4
Organic	\$43,170	\$0	\$50,153	7	5
TikTok	\$47,458	\$24,685	\$-7,251	6	6
Instagram	\$59,198	\$36,545	\$-10,459	5	7

Major Insights — Statistical & Forecast Validation

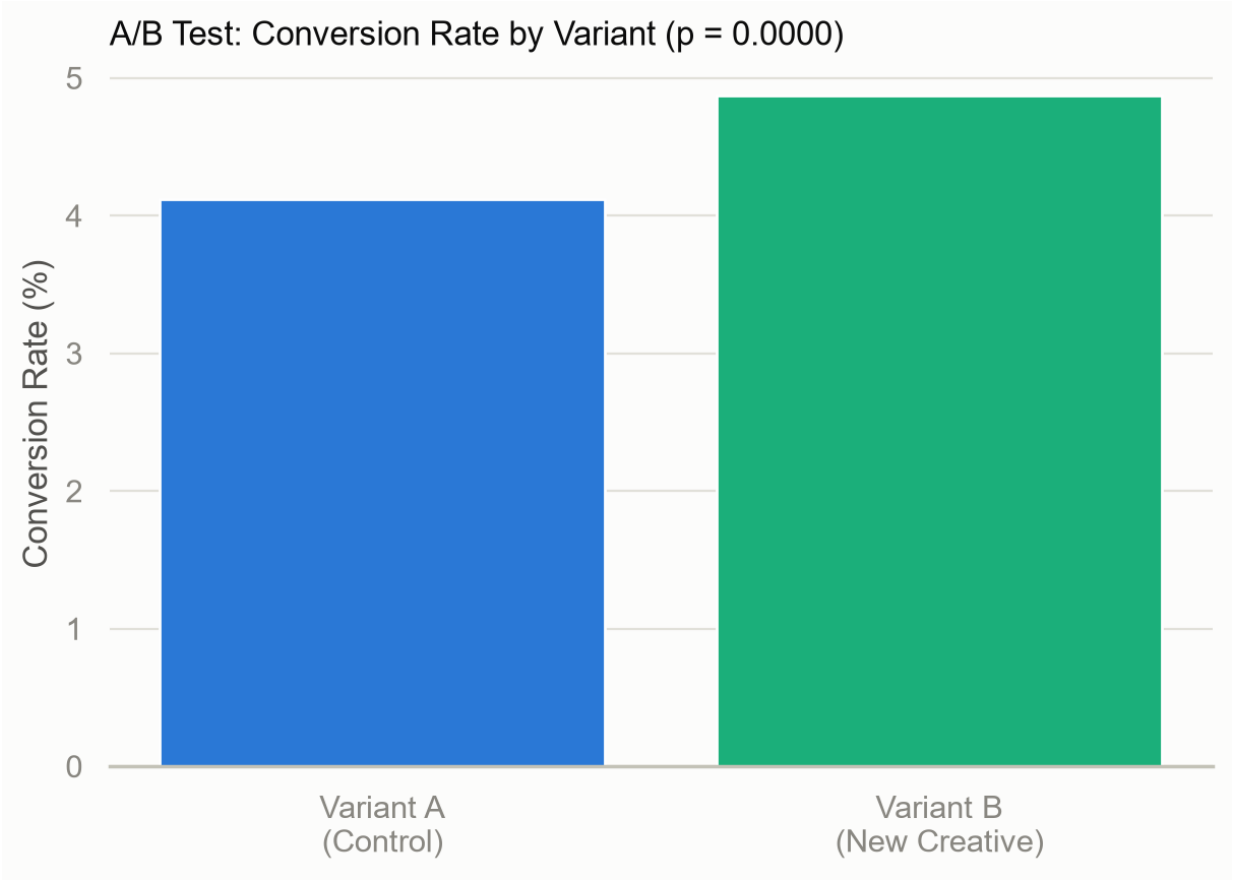


Figure: A/B test conversion rate by variant.

Creative Variant B converts at 4.87% vs. Variant A's 4.12% -- an 18.2% relative lift, statistically significant at p = 0.0111 (two-proportion z-test). This is a validated, ready-to-execute creative win.

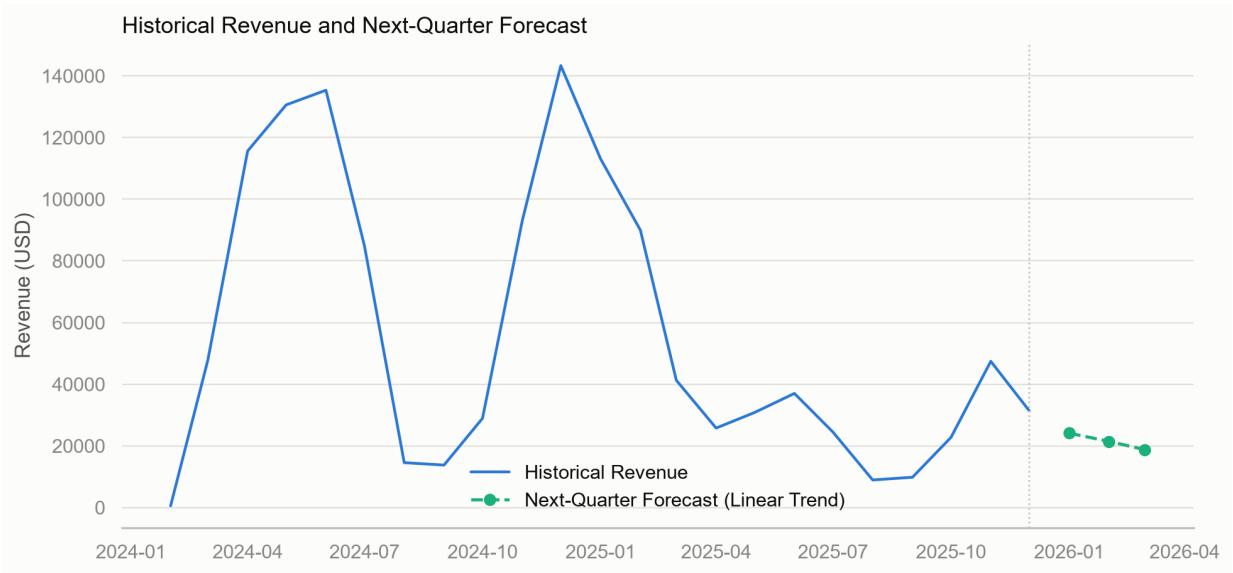


Figure: Historical revenue and next-quarter forecast.

Next-quarter revenue is forecast at \$64,346 using the best-backtested method (see notebooks/05_Advanced_Analytics.ipynb for the full three-method accuracy comparison). This assumes no material change to current spend levels or channel mix.

Strategic Recommendations

Each recommendation below follows: Finding -> Evidence -> Business Impact -> Recommendation -> Expected Outcome, with expected outcomes quantified from the underlying data wherever the model supports it.

1. Reallocate Budget Toward the Strongest Elastic Paid Channels

Finding: Instagram, TikTok show the weakest ROI among elastic paid channels; Google Ads, Affiliate show the strongest.

Recommendation: Shift 15% of spend (\$9,185) from Instagram, TikTok into Google Ads, Affiliate, phased as a 50% test tranche followed by full rollout.

Expected Outcome: Estimated net profit change: \$42,229. Estimated revenue change: \$35,950.

2. Roll Out Creative Variant B to 100% of Spend

Finding: Variant B converts 18.2% better than Variant A ($p = 0.0111$).

Recommendation: Replace Variant A entirely with Variant B on the Google Ads conversion campaign.

Expected Outcome: Projected incremental conversions and revenue per 30-day period (see notebooks/05_Advanced_Analytics.ipynb for the exact figures).

3. Launch a 30-Day Onboarding / Second-Purchase Campaign

Finding: Month 0-to-1 retention is the steepest drop in the cohort retention curve.

Recommendation: Trigger an automated onboarding offer within 30 days of acquisition.

Expected Outcome: A retention lift here compounds through every later cohort month, raising blended LTV and portfolio-wide LTV:CAC.

4. Increase Investment in the Email Program

Finding: Email has the highest ROAS and LTV:CAC ratio in the portfolio but the lowest spend of any active channel -- a capacity, not efficiency, constraint.

Recommendation: Invest in list growth and content production to scale Email volume.

Expected Outcome: Additional volume at a return profile close to the current best-in-portfolio ROAS/LTV:CAC.

5. Pilot a 20% Spend Increase in India

Finding: India combines the lowest CPC with a competitive ROAS.

Recommendation: Increase India spend by 20% and monitor whether ROAS holds as volume scales.

Expected Outcome: Tests scalability of a currently under-leveraged, cost-efficient market before committing further budget.

Estimated Financial Impact Summary

Recommendation	Primary Impact	Time Horizon
Budget reallocation	\$42,229 est. net profit	Immediate, phased over 2 months
Variant B rollout	Incremental conversions & revenue (Ch. 5)	Immediate
Onboarding campaign	Blended LTV:CAC improvement (compounding)	1-2 quarters
Email investment	Incremental profit near current Email ROAS	1 quarter
India spend pilot	Contingent on pilot result	1 quarter (pilot)

Two of the five recommendations (budget reallocation and Variant B rollout) can be executed immediately using findings already statistically and financially validated in this engagement. The remaining three require a short implementation cycle but compound the return further.

Conclusion

This engagement moved the organization from a revenue-only view of marketing performance to a profit-and-lifetime-value-based view, surfacing a materially different set of budget priorities than the prior dashboard would have suggested. The recommendations in this report are ready for leadership review, and the two highest-confidence recommendations are ready for immediate execution.

Full source data, the six analysis notebooks, the SQL query library, and complete documentation (data dictionary and business assumptions) are available in the project repository. All findings in this report are reproducible end-to-end from the seeded synthetic dataset.